JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 99, NO. B2, PAGES 2909–2924, FEBRUARY 10, 1994

An Early Carboniferous paleomagnetic pole for Gondwanaland: New results from the Mount Eclipse Sandstone in the Ngalia Basin, central Australia

Zhong Chen, Zheng-Xiang Li, Christopher M. Powell, and Basil E. Balme
Department of Geology, University of Western Australia, Nedlands

Abstract. Paleomagnetism of fourteen diamond-drill cores from the uppermost Devonian-Lower Carboniferous Mount Eclipse Sandstone in the Ngalia Basin, central Australia, has been studied. The samples were oriented using the borehole directions, bedding planes identified in the drill cores, and regional bedding attitudes. Three magnetic remanent components have been revealed: (1) a drilling-induced, low- to moderate-temperature remanence (C1) from all the drill cores, which is oriented parallel to the drill cores and is most likely a combination of isothermal remanent magnetization and piezoremanent magnetization; (2) a moderate- to high-temperature, syn-deformational overprint (C2: D = 226.7°, I = 82.3°, α_{95} = 6.2°, k = 68.9) in 65 samples from nine drill cores, which was probably acquired during the mid- to Late Carboniferous Mount Eclipse movement; and (3) a moderate- to high-temperature, likely primary remanence (C3: D = 057.1°, I = -36.9°, α_{95} = 9.6°, K = 29.6) from 64 samples in nine drill cores, which exhibits a positive fold test. C2 and C3 are found only in reddish siltstones and sandstones, whereas C1 is best developed in grey reduced sandstones and siltstones. A palynological examination gives an early to mid-Viséan (Early Carboniferous) age of deposition for samples carrying the interpreted primary component. C2 gives a mid- to Late Carboniferous pole at 32.1°S, 119.5°E with A_{95} = 11.9°, which is in agreement with results of a previous study on surface outcrops. C3, when combined with results from one drill core in a previous study, gives a Viséan pole at 37.6°S, 52.6°E with A_{95} = 8.7°. This pole suggests a south polar position for central Africa during the Early Carboniferous and therefore indicates the existence of a large ocean between Gondwanaland and Laurussia at that time. The paleo-south pole shifted from central Africa in the Early Carboniferous (~340 ±5 Ma) to East Antarctica in the mid- to Late Carboniferous (~310 ±10 Ma) with a minimal continental drifting rate of 15-20 cm/yr.

Introduction

The most significant tectonic event during the Late Paleozoic involved the formation of the supercontinent Pangea by the collision of the northern supercontinent (Laurussia) with its southern counterpart (Gondwanaland). There are two major schools of thought about when Pangea formed. One consists of researchers in paleobiogeography [e.g., *Young*, 1990] and paleoclimate [e.g., *Scotese and McKerrow*, 1990; *Scotese and Barrett*, 1990], who argue that the two supercontinents were in close proximity during the Late Devonian. The other comprises paleomagnetists [e.g., *Hurley and Van der Voo*, 1987; *Li et al.*, 1988; *Kent and Van der Voo*, 1990] and a few paleobiogeographers [e.g., *Colbath*, 1990], who argue that a large ocean still existed between Gondwanaland and Laurussia at the end of Devonian. The latter opinion is further supported by a recent paleomagnetic study in the uppermost Devonian sedimentary rocks in central Australia [*Chen et al.*, 1993]. The spread of the Late Devonian paleopoles from Australia, one of which (i.e., the Worange Point Formation [*Thrupp et al.*, 1991]) could be used to support the former opinion, has been interpreted as representing a phase of rapid apparent polar wander [*Chen et al.*, 1993; *Li et al.*, 1993; *Meert et al.*, 1993]. There is nevertheless still lack of good paleomagnetic control on the paleolatitude of Gondwanaland during the Early Carboniferous and therefore the details of its convergence with Laurussia.

To refine the Carboniferous apparent polar wander path (APWP) for Gondwanaland, *Li et al.* [1989, 1991] studied the paleomagnetism of the uppermost Devonian–Lower Carboniferous Mount Eclipse Sandstone in the Ngalia Basin, central Australia. Their samples were from both outcrops at the northern margin of the basin [*Li et al.*, 1989] and industrial diamond-drill cores in the southeastern part of the basin [*Li et al.*, 1991]. Although a mid-Carboniferous syndeformational magnetic overprint was revealed from both the surface and diamond-drill core samples, a possible primary remanence was found only in red siltstone from one drill core [*Li et al.*, 1991]. Li et al. concluded that the primary magnetic remanence in the surface outcrops and the majority of the drill-core samples (which were mainly coarse-grained, either oxidized or reduced sandstones) was overprinted by processes such as surface weathering, granular-strain at the northern margin of the basin, synorogenic chemical events and drill-barrel-induced magnetization during industrial diamond drilling. However, they speculated that the less permeable fine-grained red siltstones in the upper part of the succession, which are mostly in the subsurface in the less deformed southern part of the basin, could still retain their primary remanence. The present study is thus targeted at diamond-drill cores from the upper part of the Mount Eclipse Sandstone.

Copyright 1994 by the American Geophysical Union.

Paper number 93JB02215.
0148-0227/94/93JB-02215$05.00

Geological Setting and Palynological Analyses

Geological Setting

The Ngalia Basin is an asymmetrically faulted structural basin in the Northern Territory, central Australia (Figure 1a). The Mount Eclipse Sandstone, which is the youngest Paleozoic sedimentary unit in the basin, comprises 2400-3700 m of predominantly coarse-grained arkosic sandstone with prominent interbeds of pebble to boulder conglomerate and subordinate siltstone and shale. It was formed mainly in fluvial and piedmont environments [*Wells and Moss*, 1983].

The age of the Mount Eclipse Sandstone ranges from the latest Devonian to possibly the Late Carboniferous. Plant fragments from the basal part indicate a Late Devonian to Early Carboniferous age [*Wells and Moss*, 1983]. Samples also from the basal part yielded a late Famennian spore assemblage (D. Burger and E. M. Kemp, as cited by *Wells and Moss* [1983]). A rich spore assemblage recognized in the middle part of this formation is probably Viséan in age and belongs to the *Anapiculatisporites largus* Assemblage (D. Burger and E. M. Kemp, as cited by *Wells and Moss* [1983]). G. Playford, as cited by *Wells and Moss* [1983], has confirmed the above identification and reported a similar flora in a sample from the DAV1 drill core. *Kemp et al.* [1977] pointed out that this Mount Eclipse palynoflora is of mid-Early Carboniferous (Viséan) age. Although the control on plant macrofossil ages is not tight, plant fragments from the upper part of the formation in Djambidjimba Mesa are possibly of Late Carboniferous age [*Wells and Moss*, 1983]. Therefore the Mount Eclipse Sandstone is considered to have been deposited during late Famennian to possibly mid-Carboniferous times.

The sedimentary succession in the Ngalia Basin was deformed by a series of regional movements. The Mount Eclipse movement, a local name for the more regional Alice Springs Orogeny (ASO), was the strongest deformational event in the Paleozoic history of this basin and adjacent areas [*Wells and Moss*, 1983; *Shaw and Black*, 1991]. Both the Paleozoic sedimentary succession and the Precambrian basement were deformed during the ASO [*Wells and Moss*, 1983]. The deformation of the sediments is most intense along the northern margin of the basin where the strata are locally overturned. The intensity of deformation decreases to the south [*Wells and Moss*, 1983]. Based on Rb-Sr age determinations of the major metamorphic events in the Arunta Block, south of this basin [*Mortimer et al.*, 1987] (older data

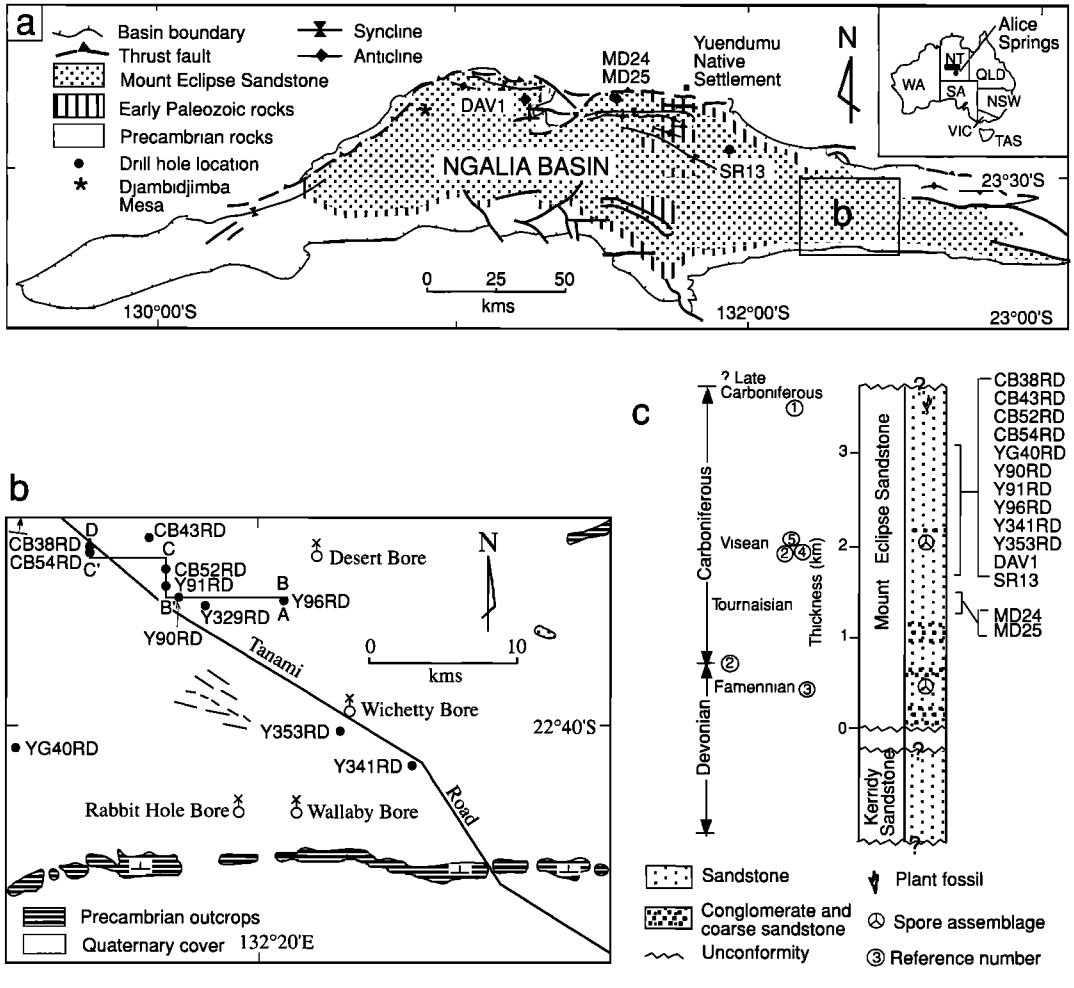


Figure 1. (a) Regional geology and (b) location of the drill holes in the Ngalia Basin and (c) stratigraphic positions of the paleomagnetic samples. Sources for age assignations in Figure 1c are 1, Appendix 6 of *Wells and Moss* [1983]; 2, R. Evans et al. (unpublished report) as cited by *Saucier* [1981]; 3, D. G. Morris (personal communication, 1985); 4, G. Playford as cited by *Wells and Moss* [1983]; 5, this study.

summarized by *Powell and Veevers* [1987]), *Li et al.* [1989] suggested that the age of the Mount Eclipse movement is about 320 ±10 Ma.

A total of 283 core segments, each about 10-20 cm long, was collected from 14 diamond-drill cores stored in the Core Library of the Northern Territory Geological Survey, located at Alice Springs, central Australia. Eleven of the diamond-drill cores are 36 mm in diameter, and three (DAV1, MD24, and MD25) are 48 mm in diameter. These cores were drilled by exploration companies during stratigraphic drilling and uranium exploration programmes in the Ngalia Basin. Paleomagnetic sampling was concentrated on the reddish medium- to fine-grained materials as these samples were considered to have a better chance of retaining their primary remanence [*Li et al.*, 1991]. Nevertheless, some samples were collected from grey (reduced) rocks for comparison.

New Palynological Analysis

In order to identify the stratigraphic positions of the selected core segments, 13 samples were collected from the drill cores for palynological (spore) analysis. All the samples were prepared using HF and gravity separation with $ZnCl_2$. However, only six of them, each from a different core, carried enough spores to define their stratigraphic position. Results of this analysis are discussed below.

Three spore assemblages from samples CB38RD-106.60, CB52RD-290.67, and SR13-96.25 (the real number after the dash in a sample name stands for the depths in meters of the top end of that sample) were essentially similar, and there are no obvious criteria to differentiate them biostratigaphically. They are therefore regarded as more or less coeval. They contain no unambiguous marine micro-organisms, and all vascular plant spores are trilete. The dominant species is the small, probable lycopsid, spore *Granulatisporites frustulentus* Balme & Hassell, which ranges from Upper Devonian to Viséan but is very abundant in the Australian Lower Carboniferous. No Upper Devonian index forms were present, and all elements of the assemblages, including species of *Verrucosisporites, Reticulatisporites, "Cirratriradites", Convolutispora, Retusotriletes, Apiculatisporis, Velamisporites, Endosporites, Anapiculatisporites, Punctatisporites, Lophozonotriletes, Raistrickia, Crassispora and Dictyotriletes*, are typical of the Carboniferous. Moreover, in Australia, all these are more characteristic of the Viséan than the Tournaisian.

The above three assemblages are similar to previously described Viséan assemblages from the Ngalia Basin [*Playford*, 1985] and Drummond Basin [*Playford*, 1978]. This implies that these samples are Viséan, and there is slight, mainly negative due to the absence of certain species, evidence that favours an early to mid-Viséan age. The palynological data from samples Y90RD-240.03, Y341RD-277.80, and Y353RD-227.99 are very similar to those from cores CB38RD, CB52RD, and SR13 but not as conclusive.

The stratigraphic relationships between the spore assemblages identified and samples which were later shown to carry a primary remanence are illustrated in Figure 2. Although there is no more than one spore assemblage identified from any single drill core, from their regional stratigraphic correlation we can still conclude fairly confidently that the paleomagnetic samples are of Viséan, and mostly early to mid-Viséan, age. Core Y329RD of *Li et al.* [1991] was drilled from a similar stratigraphic section to core Y90RD of this study (Figure 1b) and is therefore treated as being of comparable age.

Cores MD24 and MD25 were drilled in the lower part of the Mount Eclipse Sandstone [*Chuck*, 1980] but above the Devonian/Carboniferous boundary [*Wells and Moss*, 1983; *Li et al.*, 1989; D. G. Morris, personal communication, 1985] (Figures 1a and 1c). The age of this stratigraphic section is thus believed to be earliest Carboniferous.

Paleomagnetic and Rock Magnetic Techniques

All the drill cores were described geologically before paleomagnetic sampling. Each sample was redrilled parallel to the long axis of the core, using a diamond drill bit of 25 mm inner diameter. Two or three 22-mm-long standard specimens were then sliced from the subsamples.

The diamond-drill cores are mostly shorter than 300 m, with the exception of core DAV1 (Table 1). The dip azimuths of the drilled rock strata (Table 1) were obtained from either interpretation of surrounding outcrops, or correlation of closely spaced drill cores (B. G. Jones, written communication, 1987, and unpublished reports for Agip Australia Pty Ltd). The dips of the strata were cross-checked by comparing the results of core correlation with those calculated using the bedding-to-core angles, the core orientations and the regional bedding attitudes. Errors associated with this bedding-attitude estimation are considered to be within ± 5°.

Three of the 14 diamond-drill cores (Y353RD, MD24, and MD25) were drilled obliquely, and seven (CB52RD, CB54RD, Y90RD, Y91RD, Y96RD, Y341RD, and SR13) were drilled vertically. The orientations of four cores (CB38RD, CB43RD, YG40RD, and DAV1) were not clearly stated on company reports and have been assumed to be vertical because this is the attitude at which most company drilling was carried out. This assumption is supported by the agreement between the characteristic remanent directions in these cores and those in other cores as documented later. Orientation of a core sample was achieved by first restoring the core segment to its original attitude, and then rotating the core around its long axis so to align the dip azimuth of the bedding on the core with that of the estimated regional bedding.

The anisotropy of low-field magnetic susceptibility (AMS) of most of the specimens was measured using either a modified Digico magnetic anisotropy delineator in the CSIRO Rock Magnetism Laboratory in Sydney, or a self-developed AMS system for the Bartington MS2 meter in the University of Western Australia (UWA) Paleomagnetism Laboratory. AMS of some of the thermally treated specimens was also measured at UWA.

A total of 551 specimens was subjected to stepwise demagnetization experiments. Thirteen to eighteen steps of demagnetization and measurement were carried out on half of the specimens, and 7-12 steps were applied to the rest. Both thermal and alternating field (AF) demagnetization methods [*McElhinny*, 1973; *Collinson*, 1983] were employed. As most samples are from red sandstones and siltstones, in which hematite is believed to be the main magnetic carrier, thermal demagnetization is the most efficient method. Specimens were heated in either a non magnetic automatic furnace installed within magnetic-free space (< ± 5nT) [*Coward et al.*, 1985] at CSIRO, or a modified Schonstedt thermal

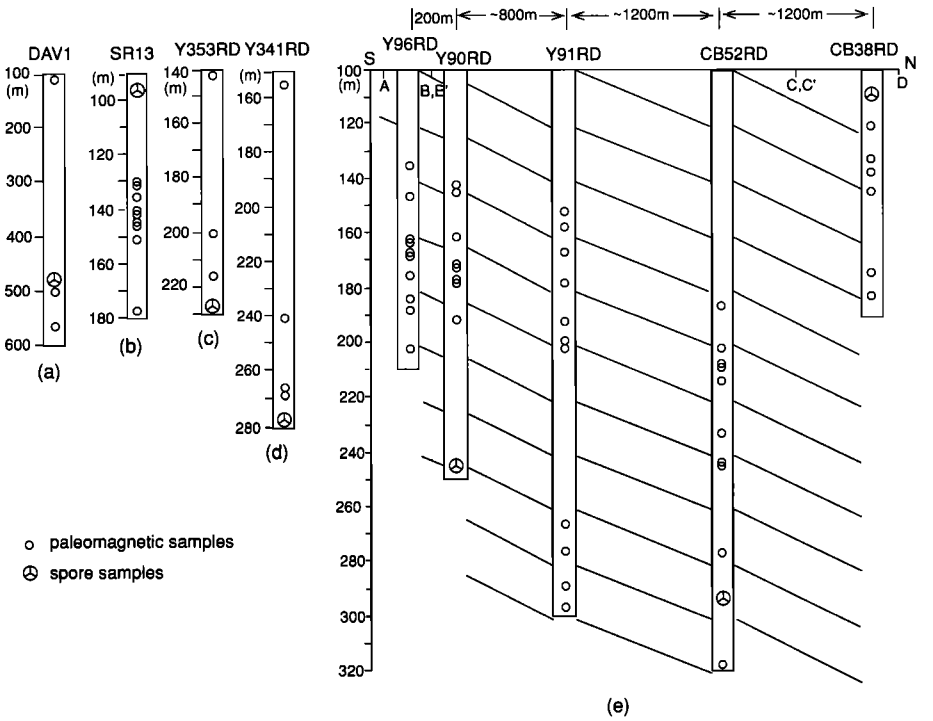


Figure 2. Stratigraphic relationship between spore assemblages and samples carrying a primary remanence: (a) core DAV1, (b) core SR13, (c) Y353RD, (d) core Y341RD, (e) a cross section of A–B, B'–C, C'–D in Figure 1b. Connecting lines are inferred correlations between adjacent drill holes.

Table 1. Information About the Studied Diamond Drill Cores

Core Name	Location		Bottom Depth (Depth Range), m	Core Orientation (Plunge/Plunge-Azimuth)	Estimated Regional Bedding, deg	Number of Samples
	°S	°E				
CB38RD	22.56	132.22	194.8	vertical*	28.0 / 009.0	28
CB43RD	22.55	132.26	203.4	vertical*	22.0 / 006.0	21
CB52RD	22.57	132.29	321.0	vertical	22.0 / 000.0	27
CB54RD	22.56	132.22	285.0	vertical	36.0 / 005.0	7
YG40RD	22.68	132.17	237.0	vertical*	15.0 / 347.0	13
Y90RD	22.59	132.28	278.3	vertical		
			(89.0 - 190.0)		20.0 / 004.0	23
			(190.0 - 278.3)		21.0 / 004.0	12
Y91RD	22.58	132.29	303.4	vertical	17.5 / 005.0	27
Y96RD	22.59	132.35	211.7	vertical		
			(122.0 - 189.0)		12.0 / 032.0	12
			(189.0 - 211.7)		15.0 / 032.0	9
Y341RD	22.69	132.44	296.0	vertical	20.0 / 320.0	21
Y353RD	22.67	132.39	237.8	(65.0 / 0.0)	73.0 / 195.0	9
DAV1	22.30	131.25	904.6	vertical*	12.0 / 313.0	18
SR13	22.39	131.92	189.0	vertical		
			(78.0 - 140.0)		28.0 / 021.0	10
			(140.0 - 189.0)		26.0 / 021.0	13
MD24	22.29	131.51	299.3			
			(179.0 - 242.0)	(63.0 / 0.0)	33.0 / 164.0	8
			(242.0 - 278.0)	(64.0 / 0.0)	36.1 / 164.0	
			(279.0 - 299.3)	(65.0 / 0.0)	38.7 / 164.0	6
MD25	22.29	131.53	235.2	(60.0 / 023.0)	23.4 / 213.0	19

* Core orientations are not clearly stated on the company reports but are assumed to be vertical (see discussion in text).

demagnetizer at UWA. They were heated for about 45 to 60 min and then cooled by compressed air. Two-layer μ-metal containers were used to transport the specimens from the furnaces to the magnetometers. A Schonstedt single-axis AF demagnetizer at CSIRO, or a Molspin two-axis tumbler AF demagnetizer at UWA was used for AF demagnetization on some representative samples. After AF demagnetization of up to 100-mT peak field, thermal demagnetization was usually required to demagnetize the specimens totally. Magnetic remanences were measured either with a three-axis CTF cryogenic magnetometer at CSIRO (instrument noise level ~2 x 10^{-9} mA/m), or an upgraded Digico spinner magnetometer at UWA (instrument noise level 3-5 x 10^{-8} mA/m). A three-axis Helmholz coil system is installed around the cryogenic magnetometer in CSIRO, and a single-axis inclined Helmholz coil system [*Collinson*, 1983] is installed around all magnetic equipment at UWA to reduce the influence of the present geomagnetic field on the specimens. Both magnetometers were driven by the same CSIRO ROCMAG software package.

Data analysis was carried out by utilizing *Zijderveld* [1967] diagrams, principal component analysis [*Kirschvink*, 1980], and stereographic projections. Linearity (the ratio of the distance between the bounding data points to the sum of the distance between the intervening points [*Schmidt*, 1982]) was used as the parameter for identifying components of magnetization in each specimen. Statistics for mean-direction calculation follow *Fisher* [1953].

Each diamond-drill core in this study has been treated as a site, and each core segment has been treated as a sample. Remanences from different samples were generally compared with each other on the basis of their unblocking temperatures (or their coercivities in the case of AF demagnetization). The fold test statistics described by *McFadden* [1990], based on a test statistic that is sensitive to correlation between the distribution of site-mean directions about the overall mean direction and the tectonic corrections, were adopted in this study.

Results

Anisotropy of Low-field Magnetic Susceptibility (AMS)

The anisotropy of low-field magnetic susceptibility (AMS) for a total of 342 specimens from 14 diamond-drill cores was measured. The purpose of these measurements was to check (1) if the sampled strata have been subject to significant tectonic strain; and (2) whether the bedding plane on each core segment, which is the key feature employed for sample orientation, was accurately identified.

The AMS of cores DAV1, YG40RD, Y91RD, Y341RD, and Y353RD measured on a Digico anisotropy delineator at CSIRO before any thermal treatment gives rather simple results. The K_{min} (minimum principal axis) directions of specimens from each core are well grouped, and their mean is not significantly different from the pole to the regional bedding for that core. The K_{max} (maximum principal axis) and K_{int} (intermediate principal axis) directions lie on the bedding plane (e.g., Figures 3a and 3b), demonstrating a compactional magnetic fabric [e.g., *Kligfield et al.*, 1983].

The AMS based on six-position measurements on the Bartington MS2 meter is somewhat more complicated. The K_{min} directions for seven out of the nine cores (CB38RD, CB52RD, Y90RD, Y96RD, MD24, MD25, and SR13) are reasonably well grouped, but their mean directions depart significantly from the poles of regional bedding (e.g., Figures 3c and 3g). Data from cores CB43RD and CB54RD are dispersed (e.g., Figure 3e). However, after heating steps between 550°C and 650°C, the magnetic susceptibility and anisotropy of all these specimens were enhanced, the K_{min} directions for all nine cores became better grouped, and their means fell close to the poles of regional bedding (Figures 3d, 3f, and 3h). That is, the magnetic foliation (the plane containing the K_{max} and K_{int} axes) became close to parallel with the bedding planes, agreeing with the results measured on a Digico anisotropy delineator. Isothermal remanent magnetization (IRM) measurements, differential thermal (DTA) and thermal gravimetric analyses (TGA), X ray diffraction analysis, Leco-induction-furnace analysis and Mössbauer analysis indicate that it was probably the thermal conversion of pyrrhotite and other Fe-sulfides into magnetite that caused the enhancement of magnetic susceptibility and anisotropy. The discordant magnetic fabric measured on the MS2 meter before thermal trement is believed to be erroneous due to the inadequate number of measurement steps. Details of these experiments and the effects of heating on magnetic mineralogy and magnetic fabric will be presented elsewhere.

The parallelism of observed bedding with magnetic foliation in almost all the samples, either measured on the Digico system before heating or on the Bartington MS2 meter after heating, indicates that our identification of bedding in the core is reasonably accurate. If we had mistaken cross bedding for the regional bedding, the compactional magnetic foliation [e.g., *Kligfield et al.*, 1983] would have been oblique. The well-preserved bedding-parallel foliation also indicates that these samples have not undergone significant internal strain.

Paleomagnetic Results

The intensities of natural remanent magnetization (NRM) range predominantly from 1 to 8 mA/m (x10^{-6} emu/cm^3). Three characteristic remanent magnetization components have been identified. The first (C1) is a steep upward component in the specimen coordinate system, with low to moderate unblocking temperatures. The second (C2) is a steep downward component in the geographic coordinate system, with moderate to high unblocking temperatures. The third (C3) is generally a shallow upward to northeast, or downward to southwest, component in the stratigraphic coordinate system with high unblocking temperatures. In two of the 14 cores (MD24 and MD25), there is no stable component which meets the criterion of three or more samples containing the same component and with an α_{95} around the resulting mean ≤ 30°. Orthogonal plots of some typical demagnetization data are shown in Figure 4. Discussion and interpretation of the three components follow:

Drilling-induced remanence (DIR): (C1). C1 is dominant in the reduced (grey) rocks but less so in the oxidized (reddish) samples. It can generally be eliminated either by AF demagnetization with a peak field between 10 and 75 mT (100 and 750 Oe) (Figure 4k) or by low-temperature (200°C-400°C) thermal demagnetization (Figures 4a-4c and 4h). However, the unblocking temperature can extend up to 500°C or even higher in some specimens (Figures 4i and 4j).

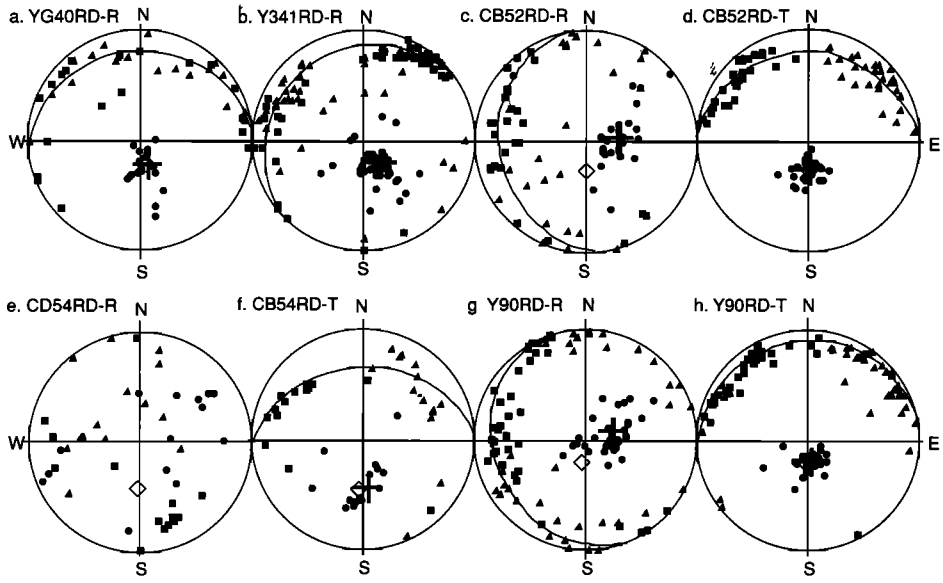


Figure 3. Lower hemisphere equal-area stereographic projections of the anisotropy of magnetic susceptibility (AMS) data. Squares represent directions of the maximum axes (K_{max}) of specimens; triangles represent the intermediate axes (K_{int}); dots represent the minimum axes (K_{min}); crosses represent the means of K_{min}, and diamonds represent the poles to the regional bedding. Great circles are the planes of the magnetic foliation. "-R" after the drill core names represents AMS data before any thermal treatment, and "-T" represents data after 550°C heating.

In reduced specimens, usually only C1 can be found, beyond which the remanent vector showed irregular behavior (Figure 4i). Thermal demagnetization of isothermal remanent magnetization (IRM) of two typical grey specimens with C1 (Figure 4i and specimens Y341RD-203.8 and Y90RD-240.2 in Figure 5) shows that the majority of the IRM was carried by a mineral phase with a Curie temperature between 300°C and 350°C, indicating pyrrhotite is the major magnetic phase of C1.

C1 was found in 66% of the samples from 11 diamond-drill cores (Table 2 and Figure 6) after principal component analysis with a linearity cutoff level of 0.98. The core-mean directions are significantly oblique to the present-day local magnetic field and dipole field directions. They group very well in core coordinates (Figure 6l), become slightly more dispersed in geographic coordinates (Figure 6m), and significantly more dispersed in stratigraphic coordinates (Figure 6n and Table 2). Similar results were reported by *Li et al.* [1991] and were interpreted to reflect a drilling-induced remanence (DIR).

Data from this study reinforce such an interpretation. The development of C1 in any diamond-drill core depends upon its core diameter and the degree of oxidation of the drilled strata. No C1 component was revealed from standard specimens redrilled from the center of the reddish segments with larger (48 mm) diameters (i.e., cores MD24, MD25, and DAV1). However, C1 was found in the reduced specimens from core DAV1 and was revealed in 10 out of the 11 cores with smaller (36 mm) diameter. C1 in reduced samples is normally better grouped and better aligned with the core axes. These phenomena confirm *Li et al.*'s [1991] conclusions that (1), hematite is more resistant to the DIR overprint than magnetite and other magnetic minerals (i.e., pyrrhotite and goethite) with lower unblocking temperatures; and (2), the more interior part of a core segment is less affected by DIR.

All core segments, except those from core SR13, were redrilled from the tops of segments during specimen preparation in laboratory. The segments from core SR13 were reversely redrilled. Sawing was done in a random manner. However, C1 revealed in the samples of SR13 (Figures 4j and 4k) has the same characteristics as that from the other cores. This rules out the possibility that C1 was acquired during specimen preparation [*Burmester*, 1977; *Jackson and Van der Voo*, 1985]. A more realistic mechanism for the DIR overprint is probably the combination of isothermal magnetization, and piezo-remanent magnetization, formed in an axially symmetric magnetic field inside the drill bit during the industrial drilling [*Kodama*, 1984; *McWilliams and Pinto*, 1988; *Audunsson and Levi*, 1989; *Li et al.*, 1991].

Mid- to Late Carboniferous syndeformational overprint: (C2). The most notable feature of component C2 is its steep downward direction in geographic coordinates and its moderate to high unblocking temperature (Figures 4b and 4h). This component is found only in reddish rock samples, and it does not always decay toward the origin in the vector plots. C2 is resistant to AF demagnetization but can be efficiently demagnetized at moderate to high temperature (300°C to 650°C). Therefore hematite is most likely its main carrier, although magnetite could also contribute. C2 can be found in only about 33% of samples from nine of the 14 diamond-drill cores after a principal component analysis with a linearity cutoff level of 0.90 (Table 3). Among the nine diamond-drill cores, one (DAV1) is located in the northern part of the basin and the rest in the southeastern part.

The core-mean directions from eight out of the nine cores group reasonably well both before and after bedding correction (Figures 7j and 7l), but one core (Y353RD) disperses away from the main group both before and after bedding correction (Figure 7l). The best grouping of the nine core means is achieved after ~50% of untilting (Figures 7k and

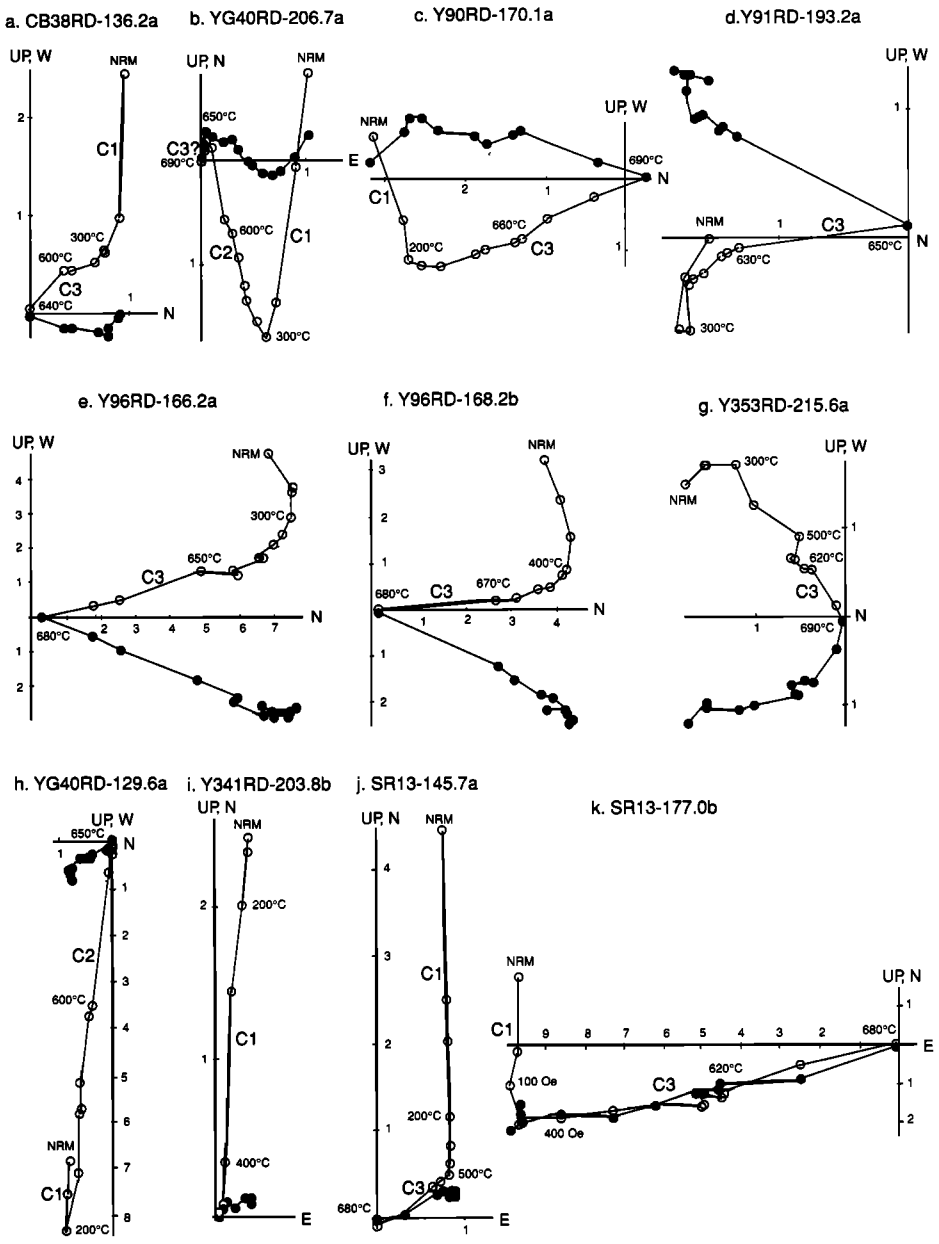


Figure 4. Representative orthogonal plots of the *in situ* magnetization vectors revealed from the Mount Eclipse Sandstone. Solid (open) circles are projections on horizontal (vertical) planes. C1, C2 and C3, are labeled on each plot. Scales are in 1 mA/m ($1x10^{-6}$ emu/cm^3).

8). The fold test (definition 1) of *McFadden* [1990] (Table 4) shows that the null hypothesis of no correlation between the magnetic directions and the bedding attitudes can be rejected with 95% confidence both before and after the bedding correction but can not be rejected after ~50% of unfolding. C2 is therefore likely to be a syndeformational overprint. Although the outcome of this fold test is largely influenced by results from core Y353RD, which was drilled from strata with a significantly different bedding attitude from the rest of the cores, the interpretation is nevertheless in agreement with previous results from both outcrop samples [*Li et al.*, 1989] and subsurface samples [*Li et al.*, 1991]. The corresponding south pole position (EL2), which is the mean of poles calculated from core-mean directions, is at (32.1°S, 119.5°E) with $A95$=11.9°, which falls close to the mid- to Late Carboniferous overprint pole (EL1) from surface samples of the northern margin of the Ngalia Basin [*Li et al.*, 1989] (Figure 10a). Note that the C2 directions from cores CB38RD, CB43RD, YG40RD, and DAV1, where vertical drill core orientation has been assumed, agree well with those of the other cores. This demonstrates that the assumed vertical orientations for these cores are reasonably accurate.

Early Carboniferous Primary Remanence: (C3). This remanence (C3) can be recognized in nine diamond-drill cores (Table 5 and Figure 9) after a principal component analysis with a linearity cutoff level of 0.90. C3 is a high-temperature component with unblocking temperature varying from 300°C up to 690°C (Figures 4a, 4c, 4d, 4e, 4f, 4g, 4j and 4k) and is found only in reddish samples. Therefore hematite is inferred to be the major magnetic carrier. The result of an

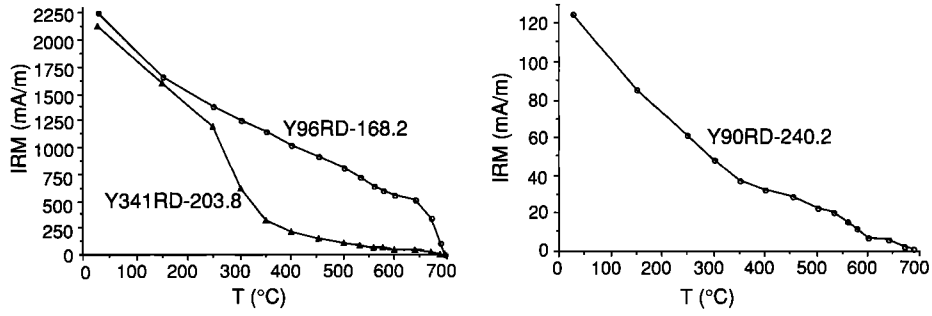


Figure 5. Thermal demagnetization curves of IRM of three typical specimens. Y96RD-168.2 is a reddish siltstone sample, and Y341RD-203.8 and Y90RD-240.2 are greyish siltstone samples.

IRM experiment on sample Y96RD-168.2 (Figure 5), which is a reddish siltstone and contains predominately the characteristic C3 (Figure 4f), supports this conclusion.

C3 commonly decays approximately towards the origin in Zijderveld plots (Figures 4a, 4c, 4d, 4e, 4f, 4g, 4j, and 4k), with its bedding-corrected directions either moderately downward to the southwest, or upward to the northeast (Figure 9k). One sample from each of cores CB52RD, Y90RD, Y96RD, and SR13 has been regarded as an outlier because their directions are significantly apart from the rest of those cores (Figures 9b, 9c, 9e and 9i).

Eight of the nine core-mean directions for C3 are from beds dipping between 15° and 28° to the north or northwest (Table 1) and are reasonably well grouped before bedding correction. The results from the ninth core, Y353RD, which has bedding dipping 73° to the south, are quite divergent (Figure 9j). After bedding correction, however, the core-mean directions from all nine cores merge (Figure 9k). A fold test, as suggested by *McFadden* [1990] (definition 1), shows that the core-mean directions of C3 have apparent correlations with their bedding attitudes before bedding correction, but there is no such apparent correlation after bedding correction (Table 4). This strongly suggests that the remanence is of prefolding origin. The mean bedding-corrected direction of C3 from the nine diamond-drill cores is -36.9°/057.1°. The corresponding south pole, which is the mean of poles calculated from core-mean directions, falls at (37.8°S, 051.8°E) with $A95 = 9.8°$.

The possible primary remanence in five samples from core Y329RD reported by *Li et al.* [1991] has the same characteristics as C3 found in this study and a similar direction. Because the age of those samples is inferred to be the same as our C3-carrying samples (see previous discussion), that result has been combined with C3 of this study. The combined data from the total of 10 cores also passed a positive fold test (Table 4). Their mean direction, -37.7°/057.3°, is not significantly different from the mean direction calculated from the nine diamond-drill cores of this study. Results from the 10 cores give a south pole at (37.6°S, 052.6°E) with $A95 = 8.7°$. The sample mean directions from all 10 cores pass a reversal test of *McFadden and McElhinny* [1990] with a "C" classification.

Notwithstanding, omitting the data from cores CB38RD

Table 2. Drilling-Induced Remanence (C1) Directions

Core Name		Drill Core Coordinates		In Situ		100% Untilting		Common	
	n(N)	*D*, deg	*I*, deg	*D*, deg	*I*, deg	*D*, deg	*I*, deg	α_{95}, deg	*k*
CB38RD	6	027.8	-80.0	216.8	-80.0	196.7	-52.9	21.1	11.1
CB52RD	19	012.0	-83.8	192.0	-83.8	182.8	-61.9	6.2	30.4
CB54RD	6	140.1	-84.6	325.1	-84.6	191.6	-58.0	11.9	32.8
SR13	21	324.3	-88.2	165.3	-88.2	198.8	-61.8	4.3	56.7
DAV1	9	045.2	-79.9	358.2	-79.9	077.2	-81.4	19.7	7.8
YG40RD	13	235.8	-87.9	042.8	-87.9	159.6	-76.1	4.0	106.4
Y90RD	19	310.8	-83.9	134.8	-83.9	172.9	-65.3	10.0	12.2
Y91RD	18	127.2	-84.1	312.2	-84.1	203.7	-75.3	11.5	9.9
Y96RD	12	122.4	-82.0	334.1	-82.0	249.9	-79.0	11.5	15.3
Y341RD	19	303.8	-89.6	083.8	-89.6	139.1	-69.8	2.7	151.2
Y353RD	6	343.3	-79.4	200.5	-72.8	013.0	-34.2	11.5	34.9
Mean	(11)	034.9	-87.3	244.9	-88.2	181.0	-76.3		
α_{95}		3.8		4.6		15.1			
k		148.1		99.9		10.1			

n (*N*) – number of samples (drill cores); α_{95}, half-angle of the 95% confidence cone; *k*, precision parameter of *Fisher* [1953] distribution.

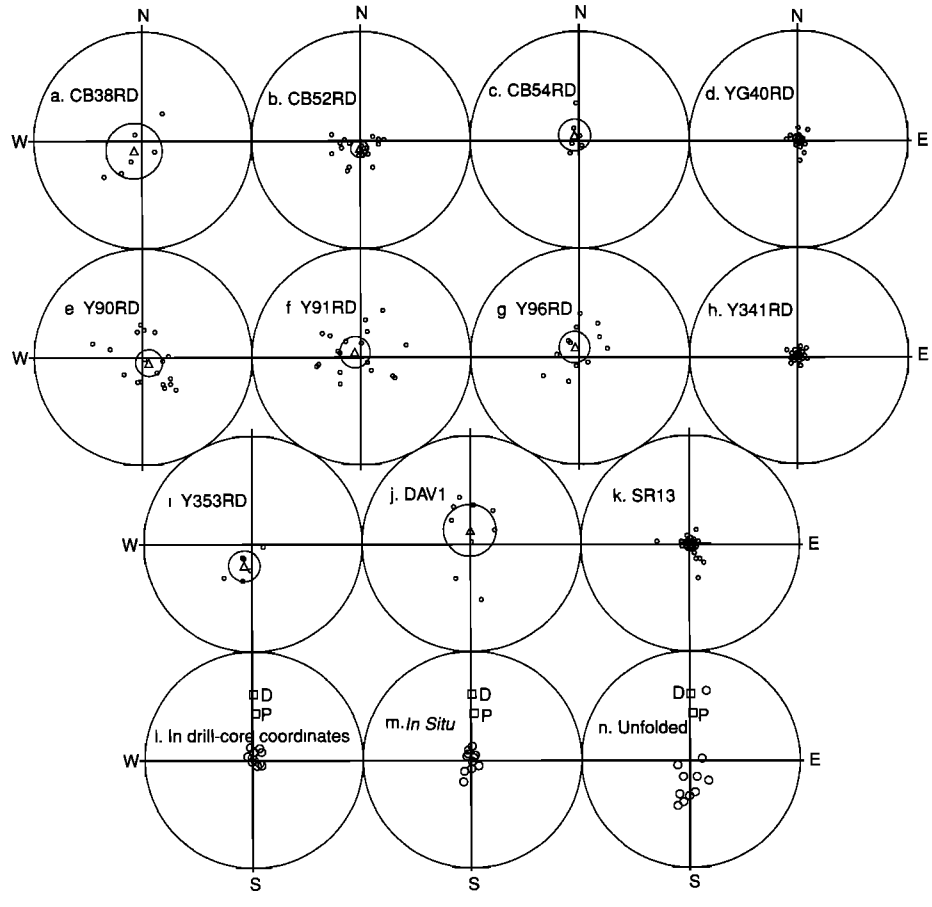


Figure 6. Upper hemisphere equal-area stereographic projections of C1 directions. (a) to (k) are in situ (geographic) sample-mean directions. Core-mean directions are shown in drill-core coordinates (l), in situ (m) and in stratigraphic coordinates (n). The means (triangles) and their $\alpha 95$ circles (small circles) are also shown in (a)–(k). P – present geomagnetic field direction, and D – present dipole field direction.

and DAV1, the orientations of which are assumed to be vertical, does not significantly alter the conclusions above or the calculated pole position (Tables 4 and 5).

Discussion and Conclusion

Summary of the New Results

The most significant result of this study is the identification of a prefolding, likely Viséan primary remanence from nine drill cores in the Ngalia Basin. This remanence exists only in oxidized fine-grained sandstone and siltstone samples. When combined with results from one core in a previous study [*Li et al.*, 1991], this remanence provides, for the first time, a reliable Viséan paleopole at (37.6°S, 52.6°E) with $\alpha 95$=8.7° for Australia and Gondwanaland and therefore has significant implications for the Gondwanan APWP and mid- to Late Paleozoic global tectonics.

A drilling-induced remanence (DIR) and a syndeformational overprint were also revealed from this study. DIR was found in 11 diamond-drill cores, but it is not as well developed as in *Li et al.'s* [1991] study on drill core materials from the same basin. DIR was identified from all samples in Li et al.'s study but from only 66% of the samples in 11 out of 14 cores in this study. Within individual cores, the mean directions of DIR were better grouped (k=100–800 and $\alpha 95$ <4°) in Li et al.'s study than in this study (k = 7–152 and $\alpha 95$ = 2.7°–21.1°) (Table 2). Possible reasons for these differences are (1) most of the specimens in *Li et al.'s* [1991] study were sliced directly from the diamond-drill core segments, whereas specimens used in this study were redrilled from the center of the core segments, where DIR has the minimum effect; (2) most of *Li et al.'s* [1991] samples were reduced sandstones and siltstones, whereas the specimens used in this study were almost all from reddish oxidised sediments, which are more resistant to DIR.

The syndeformational overprint, acquired during the latter stage of the Mount Eclipse movement, is identified in nine cores. The characteristics of this remanence are very similar to the syndeformational overprint found by *Li et al.* [1989, 1991] from both surface outcrops of the Mount Eclipse Sandstone at the northern margin of the basin and drill cores in the southern part of the basin but not as pervasive as in the surface samples. This is probably because the samples in this study are mostly from the finer-grained upper part of the Mount Eclipse Sandstone in the less deformed southeastern part of the basin and therefore less affected by the mid- to Late Carboniferous orogenic fluid due to their lower permeability [*Li et al.*, 1991] and internal strain [*Wells and Moss*, 1983].

Table 3. Mid- to Late Carboniferous Syndeformational Overprint (C2) Directions

Core Name		In Situ		50% Untilting		100% Untilting		Common	
	n(*N*)	*D*, deg	*I*, deg	*D*, deg	*I*, deg	*D*, deg	*I*, deg	α_{95}(°)	*k*
CB38RD	15	181.0	66.8	169.5	80.5	041.2	84.1	9.0	18.9
CB43RD	8	216.2	66.7	237.1	75.2	283.6	78.4	10.4	29.1
CB52RD	3	183.0	73.4	188.8	84.4	350.9	84.5	14.3	75.1
YG40RD	10	185.2	88.9	343.8	83.6	345.5	76.1	11.3	19.2
Y90RD	3	222.3	63.4	239.8	70.4	270.0	73.8	18.5	45.5
Y91RD	9	234.9	76.7	275.4	79.8	315.5	76.6	17.1	10.1
Y341RD	4	169.5	66.0	186.5	73.9	224.1	78.4	15.8	34.6
Y353RD	3	015.4	64.3	194.1	79.2	194.8	42.6	27.0	22.0
DAV1	10	301.6	88.7	311.0	82.8	311.9	76.8	7.2	46.4
Mean	(9)	201.7	79.6	226.7	82.3	266.1	82.3		
α_{95}		11.5		6.2		12.1			
k		21.1		68.9		19.0			

Mean south pole position (N =9) after 50% untilting is 32.1°S, 119.5°E; A_{95}=11.9°, K =19.7. n (N), number of samples (drill cores); α_{95}, half angle of the 95% confidence cone; A_{95}, half angle of the 95% confidence cone of the paleopole.

Late Paleozoic APWP of Gondwanaland

Figure 10 shows the revised APWP for Gondwanaland according to the poles listed in Table 6. All these poles are of C-Class or above according to the quality assessment scheme of *Li and Powell* [1993] and have a quality factor $Q \geq 3$ if assessed according to *Van der Voo's* [1990] quality scheme. Although many of these poles have poor age constraints, all have paleohorizontal control. The new pole MES from this study together with the new latest Devonian to earliest Carboniferous (late Famennian-Tournaisian) pole BC from the Brewer Conglomerate in the Amadeus Basin, central Australia [*Chen et al.*, 1993], provide important constraints on the trend of the latest Devonian to Carboniferous part of APWP (Figure 10a).

The major difference between this APWP and that revised by *Li et al.* [1990, Figure 4] and *Schmidt et al.* [1990, Figure 4] is in the mid-Carboniferous part of the APWPs. This is largely due to invoking the alternative interpretations about the ages

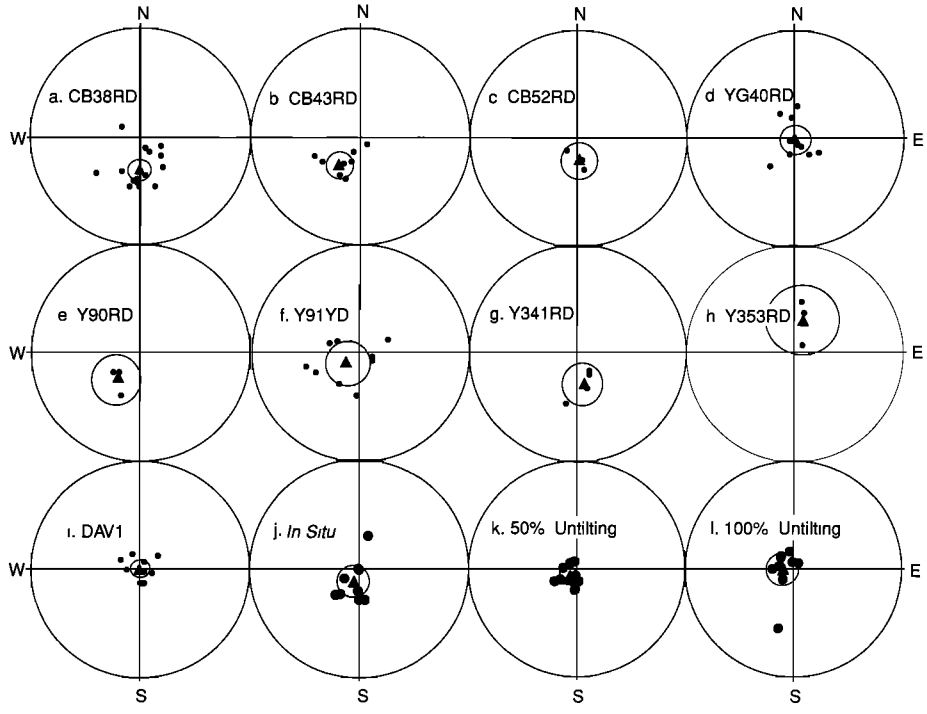


Figure 7. Lower hemisphere equal-area projections of C2 directions. (a) to (i) are in situ (geographic) sample-mean directions, (j) in situ core-mean directions, (k) after 30% of unfolding, and (l) after full tectonic correction. The means (triangles) and their α_{95} circles (small circles) are also shown.

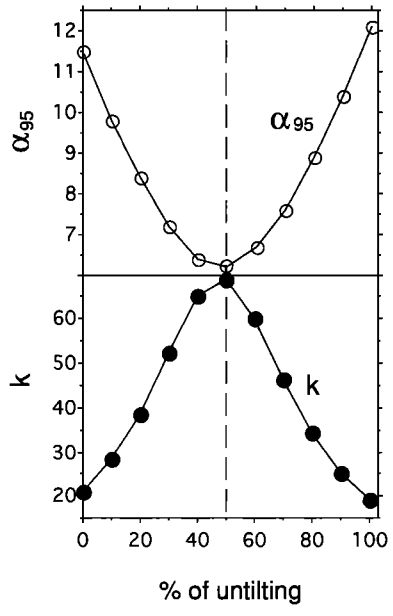


Figure 8. Changes of the Fisherian precision parameter (K) and the half angle of 95% confidence cone ($\alpha 95$) during stepwise unfolding of C2.

of poles SO, BO, DV, and GN. The two poles from southeastern Australia (SO and BO, Figure 10a), both from overprint remanence with normal polarity [*Schmidt et al.*, 1987], are more likely to be of Cretaceous age than the originally preferred mid-Carboniferous age. Pole DV (Figure 10b) from the Dwyka glacial varves in Central Africa [*McElhinny and Opdyke*, 1968] has a loose stratigraphic age constraint of Late Devonian to Late Carboniferous [*McElhinny and Opdyke*, 1968]. Although a Carboniferous age has been widely used in the past, a Late Devonian age is possible in view of the overlap of the $A95$ of pole DV with that of the coeval poles from Australia [*Li et al.*, 1993] .

Our APWP in also different from that proposed by *Bachtadse and Briden* [1991] and by *Klootwijk and Giddings* [1993]. The *Bachtadse and Briden* [1991] path was largely influenced by their results from the mid-Devonian Gilif Hills ring complexes. *Chen et al.* [1993] and *Li et al.* [1993] argued that those results could be reinterpreted within the constraint of published information to conform with the Australian data. Poles GC and BO are thus shown in Figure 10b as latest Devonian–Early Carboniferous overprint poles according to such a reinterpretation. The *Klootwijk and Giddings* [1993] path, on the other hand, was largely based on their unpublished results from Australia. The merit of that path can only be assessed upon the publication of the original results.

Our revised Carboniferous APWP agrees well with the apparent migration path of glacial centers proposed by *Caputo and Crowell* [1985] but is in sharp contrast to the APWP proposed by *Scotese and Barrett* [1990] based on paleoclimatic information. Given that the latest Devonian-Early Carboniferous segment of the Gondwanan APWP is defined by high-quality paleomagnetic poles with both paleohorizontal and age controls (see summary discussions by *Chen et al.* [1993] and *Li et al.* [1993]) (Figure 10), we suggest that the APWP of *Scotese and Barrett* [1990] is probably imprecise. The time window used by *Scotese and Barrett* [1990] for analyzing the paleoclimatic data is probably too big to be sensitive for periods of rapid continental drift.

Tectonic Significance

According to our revised APWP, the paleo-south pole "moved" from a central Africa position in the Early Carboniferous (~340 Ma) to east Antarctica in the mid- to Late Carboniferous (330-290 Ma) (Figure 10). In other words, East Gondwanaland drifted from low paleolatitude to high paleolatitude during the mid-Carboniferous (Figure 11). Accurate calibration of the drifting rate is difficult. However, if one takes the mid-Carboniferous poles (e.g., poles EL1, EL2, and KDO in Figure 10a and AEC, HB, and CS in Figure 10b) as representative Gondwanan poles with an absolute age of 320-310 Ma, it would imply a latitudinal drifting rate of 15-20 cm/yr. Although this rate appears high, it is comparable with the movement of India during the Late Cretaceous (10-15 cm/yr [*Sclater and Fischer*, 1974]), and Laurentia during the Late Precambrian [*Meert et al.*, 1993].

As a result of this rapid movement, Australia moved from a latitude range of about 10-40°S in Viséan (~340 Ma, Figure 11b) to about 50-75°S in mid- to Late Carboniferous (320-310 Ma, Figure 11c). This rapid polar-ward movement was recorded by paleontological and paleoclimatic changes. Diverse tropical or warm temperature faunas dominated in eastern and northeastern Australia during the earliest Carboniferous, but the biota suddenly became less diverse in

Table 4. Fold Test Statistical Parameters of Remanences From the Mount Eclipse Sandstone

Remanence	N	ξ_{95}	ξ_b	ξ_a	$\xi_{k\text{-max}}$	X%	ξ_{min}	Y%
C2	9	3.497	3.758	4.746	0.400	48	0.054	46
C3	9	3.497	3.803	0.642	0.051	92	0.022	91
C3*	10	3.685	4.240	1.100	0.459	91	0.027	85
C3#	8	3.298	3.506	0.498	0.123	94	0.007	92

N, number of cores; ξ_{95}, 95% confidence critical value of fold test statistic [*McFadden*, 1990]; $\xi_{b(a)}$, ξ at In Situ (after 100% of unfolding) position; $\xi_{k\text{-max}}$, ξ at maximum k position; X%, percentage of complete unfolding required for maximum k; ξ_{min}, minimum ξ; Y%, percentage of complete unfolding required for minimum ξ.

* Data taking account of the possible primary magnetic remanence obtained in this study and the data from core Y329RD by *Li et. al.* [1991].

\# Data taking account of the primary magnetic remanence obtained from both this study and *Li et al.* [1991] but omitting the cores CB38RD and DAV1, for which the core orientations were not clearly stated in the reports.

Table 5. List of the Primary Remanence (C3) Directions From the Mount Eclipse Sandstone

Core Name		In Situ		100% Untilting		Common		VGP After Untilting	
	n(*N*)	*D*, deg	*I*, deg	*D*, deg	*I*, deg	α95, deg	*k*	Latitude (°S)	Longitude (°E)
CB38RD	6	219.2	20.4	229.5	43.4	16.6	17.3	44.9	056.0
CB52RD	10	239.4	24.5	250.7	33.9	13.4	13.9	24.3	053.3
Y90RD	8	220.4	29.3	230.6	44.5	12.1	21.7	44.1	057.5
Y91RD	11	245.5	28.2	255.5	35.6	21.6	5.4	20.3	055.9
Y96RD	10	047.1	-29.9	049.7	-42.1	12.9	14.7	44.6	054.9
Y329RD*	5	042.2	-24.3	059.9	-45.0	24.8	10.4	35.9	059.6
Y341RD	4	021.5	-35.6	037.0	-42.8	22.4	17.8	56.2	053.6
Y353RD	3	118.0	-29.0	079.5	-19.3	24.3	26.7	13.4	047.7
DAV1	3	219.5	26.0	225.4	26.1	08.2	228.7	46.2	038.1
SR13	9	224.1	12.1	229.4	36.2	16.1	11.2	44.1	048.7
Mean									
Normal	32			061.2	-42.2	09.5	8.1		
Reverse	37			235.9	35.9	06.4	14.6		
Total	(10)	050.4	-28.0	057.3	-37.7				
α95		14.2		8.6					
k		12.5		32.2					
South pole (MES)								37.6	052.6
A_{95}								8.7	
K								31.7	
Mean#	(9)	051.4	-28.4	057.1	-36.9				
α95		16.0		9.6					
k		11.3		29.6					
South pole								37.8	051.8
A_{95}								9.8	
K								28.8	
Mean@	(8)	053.5	-29.1	060.0	-38.3				
α95		18.0		10.1					
k		10.4		30.7					
South pole								35.5	053.8
A_{95}								10.4	
K								29.3	

* Data reported by *Li et al.* [1991].

Mean that does not include data from core Y329RD.

@ Mean that excludes data from cores CB38RD and DAV1, for which the core orientations were not clearly stated in company reports. Other symbols are the same as in Table 3.

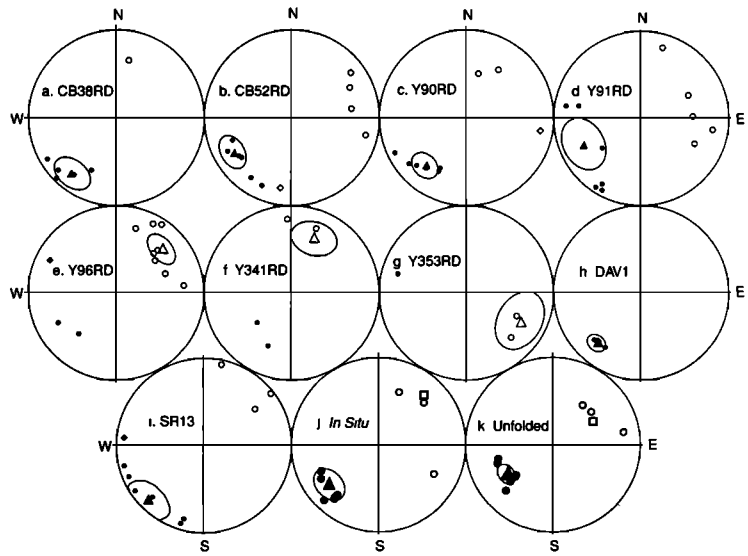


Figure 9. Equal-area stereographic projections of C3 directions. (a) to (i) are in situ (geographic) sample-mean directions. Core-mean directions are shown in both (j) geographic and (k) stratigraphic coordinates. Diamonds are the outlier sample directions excluded from mean calculations. The means (triangles) and their α95 circles (small circles) are also shown. Squares in Figures 9j and 9k are results from core Y329RD obtained by *Li et al.* [1991]. Solid (open) symbols are projections on lower (upper) hemisphere.

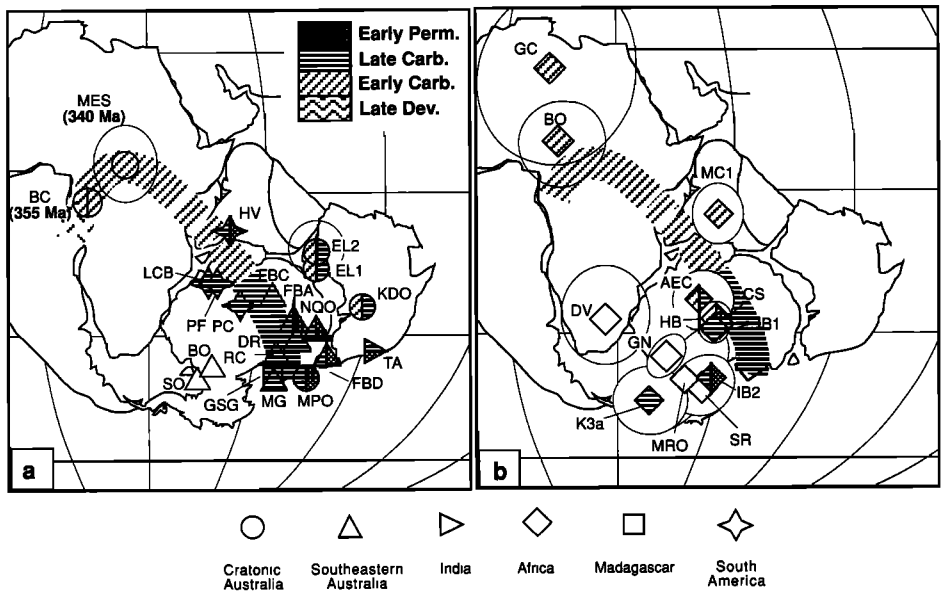


Figure 10. Revised Carboniferous APWP of Gondwanaland based on poles listed in Table 6. (a) Data from Australia, India, Madagascar and South America, and (b) data from Africa. Open diamonds in Figure 10b are poles with possible Devonian ages (Table 6) [*Li et al.*, 1993]. Gondwanaland reconstruction after *Powell et al.* [1988] and C. R. Scotese and C. M. Powell (unpublished work, 1986), which is similar to that of *Scotese and McKerrow* [1990]. Stratigraphic age assignments after *Cowie and Bassett* [1989].

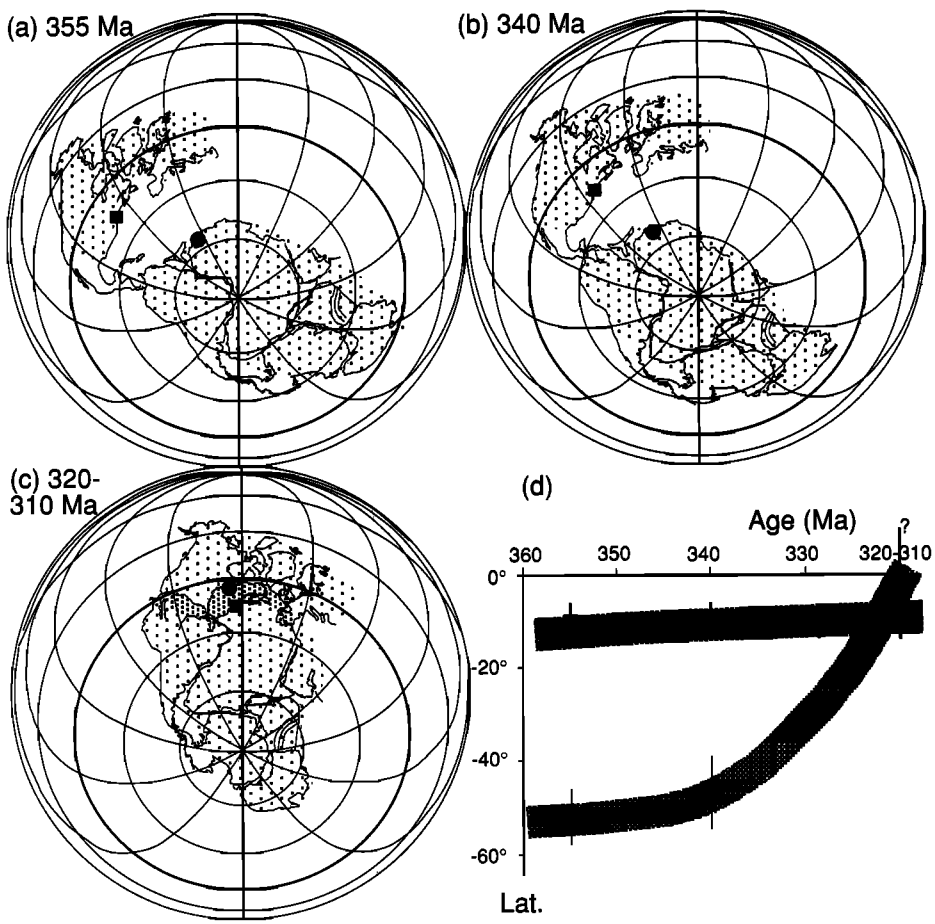


Figure 11. Relative positions of Gondwanaland and Laurentia during the Early to mid-Carboniferous based on palaeomagnetic results. Gondwanaland is plotted according to poles BC (355 Ma), MES (340 Ma), and an interpolated mid-Carboniferous pole from poles EL1, EL2, AEC, HB, and CS. Laurussia is plotted according to the averaged pole for 309–365 Ma given by *Van der Voo* [1990]. The change of the relative latitudinal distance between the two supercontinents is demonstrated in Figure 10d using one reference point from each supercontinent (dot for Gondwanaland and square for Laurussia). In Figures 10c and 10d, the apparent overlap between the two supercontinents at 320–310 Ma can be reconciled by taking into account the confidence intervals (thin vertical lines) of the paleomagnetic data into account.

Table 6. Selected Carboniferous Paleomagnetic Poles from Gondwanaland

Rock Formation	Mnemonic	Magnetization Age, Ma	Plat., °N	Plong. °E	A_{95} (DP, DM)	References
			Southeastern Australia			
Overprint (OP), NE Qld.	NQO	C2-P_1	-42.1	135.9	14.7	*after Klootwijk and Giddings* [1993]
Featherbed Volc. Field						*Klootwijk and Giddings* [1993]
(~280 Ma)	FBA	C2-P_1	-43.5	129.9	(6, 7)	
(~290 Ma)	FBD	C2-P_1	-40.1	145.4	(10, 10)	
(~305 Ma)	FBC	C2-P_1	-45.6	17.2	(9, 11)	
Dotswood redbeds	DR	C_2?	-46.1	135.6	15	*Chamalaun* [1968]
Rocky Ck Conglomerate	RC	C_2	-52.0	138.0	17	*Irving* [1966]
Main Glacial Stage	MG	C_2	-53.0	148.0	11	*Irving* [1966]
Buchan Cave Lst. (OP)	BO	mid-C, or K?	-64.7	127.9	(4.0, 4.5)	*Schmidt et al.* [1987]
Snowy River Volc. (OP)	SO	mid-C, or K?	-68.7	132.5	(5.0, 5.7)	*Schmidt et al.* [1987]
			Cratonic Australia			
Mt Painter overprint	MPO*	P-C	-45.9	149.5	12.8	*Idnurm and Heinrich* [1993]
Kulgera Dyke Swarm (OP)	KDO	mid-C	-30.0	138.0	24	*Camacho et al.* [1991]
Mt Eclipse Sst. (OP)	EL1	mid-C	-33.8	121.2	(19.2, 19.7)	*Li et al.* [1989]
Mt Eclipse Sst. (OP)	EL2	mid-C	-32.1	119.5	11.9	this study
Mt Eclipse Sst.	MES	C_1	-37.6	052.6	8.7	this study
Brewer Conglomerate	BC	D-CBoundary	-47.1	041.0	6.4	*Chen et al.* [1993]
			India			
Talchir Series	TA	C_2-P	31.5	134.3	(3.2, 2.6)	*Wensink and Klootwijk* [1968]
			Madagascar			
Lower Sakamena Group Glacial Series	GSG	C_2	-47.9	084.1	8.1	*McElhinny and Embleton* [1976]
			Africa			
K3 beds	K3a	C_2?	-45.5	040.0	8	*Opdyke* [1964] and *McElhinny and Opdyke* [1968]
Illizi Basin lst.	IB2	C_2-P_1	-38.5	057.5	2.1	*Henry et al.* [1992]
Illizi Basin lst.	IB1	C_2	-28.7	055.8	2.9	*Henry et al.* [1992]
Hassi Bachir Fmn	HB	mid-C	-26.8	056.6	3.7	*Daly and Irving* [1983]
Carboniferous sediments	CS	mid-C~C_2	-28.0	058.0		*Abou-Deeb and Tarling [1984]*
Aïn Ech Chebbi Fmn	AEC	mid-C	-22.9	051.8	6.3	*Daly and Irving* [1983]
Sabaloka ring complex	SR	D_1-C_1	-42.0	055.7	7.3	*Soffel et al.* [1990]
Mejeria red sst unit (OP)	MRO	D-P?	-40.5	049.9	(6.9, 13.5)	*Kent et al.* [1984]
C_1 results compilation	MC1	C_1	- 4.8	055.5	6.1	*Martin et al.* [1978]
Dwyka glacial varves	DV	D_3~C_2	-26.5	026.5	10.5	*McElhinny and Opdyke* [1968]
Gneiguira supergroup	GN	D_{1-2}?	-35.2	043.6	(3.0, 5.6)	*Kent et al.* [1984]
Bokkeveld Group	BO	D_{1-2} or C_1?	10.0	015.0	(8.9, 11.1)	*Bachtadse et al.* [1987]
Gilif Hills ring complexes	GC	mid-D, or C_1?	25.9	011.6	(19.6, 20.6)	*Bachtadse and Briden* [1991]
			South America			
Hoyada Verde Fmn	HV	C_2-P_1	-41.9	356.2	(8.3, 6.0)	*Rapalini and Vilas* [1991]
Piaui Fmn	PF	C_2	-50	345.0	10.4	*Creer* [1970]
Ordovician sediments at Pular and Cas Fmn	PC	C_2	-57	350.0	6	*Jesinkey et al.*, [1987]
La Colina Fmn	LCB	C_2	-49	343.0	5	*Sinito et al.* [1979]

Plat., Plong., latitude and longitude of the paleopole; A_{95}, half angle of the cone of 95% confidence [*Fisher*, 1953] around the pole; DP, DM, the semiaxes of the elliptical error around the pole at a probablity of 95%, DP is the colatitude direction and DM is perpendicular to it. Age mnemonics: D, Devonian; C, Carboniferous; P, Permian; K, Cretaceous. 1, 2, and 3 stand for Early, Middle, and Late in the Devonian; and 1 and 2 represent Early and Late in the Carboniferous and Permian.

* The mean of poles MPG(H), MPH, MPD(I), and MPS(H) of *Idnurm and Heinrich* [1993].

the mid-Carboniferous. Cool temperature provincial faunas developed in the Late Carboniferous and dominated in the Permian [*Runnegar and Campbell*, 1976]. Paleoclimatic studies [*Crowell and Frakes*, 1971; *Powell and Veevers*, 1987; *Veevers and Powell*, 1987] also show that the oldest Late Paleozoic glaciation in Australia is the late Namurian or Westphalian (mid- to Late Carboniferous) alpine glaciers in the northern Tamworth Belt, New South Wales, with the inference that continental glaciation covered much of Australia during the Late Carboniferous.

This phase of rapid continental movement ceased in Late Carboniferous and Permian time, possibly recording the collision of Gondwanaland with Laurussia, and the formation of Pangea. Figure 11 shows the paleolatitudinal relationship between the two supercontinents. It is obvious that a large ocean several thousand kilometers wide existed between Gondwanaland and Laurussia during the Late Devonian to Viséan time. This ocean was closed rapidly during the mid-Carboniferous. More high-quality mid-Carboniferous paleomagnetic poles are required in order to detail further the closure process.

Acknowledgments. This work was supported by ARC grants A38831488 and A39030719. We thank P. W. Schmidt, D. A. Clark, M. A. Lackie, and M. Huddleston, all of CSIRO Division of Exploration Geoscience, for discussions and assistance during this study, and for the use of their rock magnetism laboratory. We also thank A. Muraro, Agip Australia Limited, for permission to sample the drill cores; B. G. Jones, University of Wollongong, for information about the orientation of the strata and the drill cores, D. Johnson, C. N. G. Dampney and J. Gillings for donating a copy of their INTERMAP graphic system; A. Trench, the University of Western Australia, for comments on a draft version of this paper; M. Gaff, Library of the Department of Mines and Energy, Australian Northern Territory Geological Survey, for helping to find geological information about the research area; and E. Kingdom, the Core Library of the Northern Territory Geological Survey, for sampling assistance. Critical reviews by P. L. McFadden, C. Klootwijk, and an anonymous referee improved the manuscript. M. Corkeron typeset the final manuscript.

References

Abou-Deeb, J. M., and D. H. Tarling, Upper Palaeozoic palaeomagnetic results from Algeria and Tunisia, *Tectonophysics, 101*, 143-157, 1984.

Audunsson, H., and S. Levi, Drilling-induced remanent magnetization in basalt drill cores, *Geophys. J. Int , 98*, 613-622, 1989.

Bachtadse, V., and J. C. Briden, Palaeomagnetism of Devonian ring complexes from the Bayuda Desert, Sudan – New constraints on the apparent polar wander path for Gondwanaland, *Geophys. J. Int., 104*, 635-646, 1991.

Bachtadse, V., R. Van der Voo, and I. W. Hälbich, Paleomagnetism of the western Cape Fold Belt, South Africa, and its bearing on the Paleozoic apparent polar wander path for Gondwana, *Earth Planet Sci. Lett., 84*, 487-499, 1987.

Burmester, R. F., Origin and stability of drilling induced remanence, *Geophys. J. R. Astron. Soc., 48*, 1-14, 1977.

Camacho, A., B. Simons, and P. W. Schmidt, Geological and palaeomagnetic significance of the Kulgera Dyke Swarm, Musgrave Block, N.T., Australia, *Geophys. J. Int , 107*, 37-45, 1991.

Caputo, M. V., and J. C. Crowell, Migration of glacial centers across Gondwana during Paleozoic Era, *Geol. Soc Am Bull , 96*, 1020-1036, 1985.

Chamalaun, F. H., The magnetization of the Dotswood red beds (Queensland), *Earth Planet. Sci. Lett., 3*, 439-443, 1968.

Chen, Z., Z. X. Li, C. M. Powell, and B. E. Balme, Paleomagnetism of the Brewer Conglomerate in central Australia, and fast movement of Gondwanaland during the Late Devonian, *Geophys. J. Int., 115*, 564-574, 1993.

Chuck, R. G., Final report, on exploration in E. L. 1419, Mount Doreen, during the period 21. 04. 1977 to 20. 04. 1980, ALCOA of Aust. Limited Explor. Div., Melbourne, Victoria, Australia, 1980.

Colbath, G.K., Palaeogeography of Middle Palaeozoic organic-walled phytoplankton, in *Palaeozoic Palaeogeography and Biogeography, Mem. 12*, edited by W. S. McKerrow and C. R. Scotese, pp. 207-213, Geological Society, London, 1990.

Collinson, D. W., *Methods in Rock Magnetism and Palaeomagnetism*, 503 pp., Chapman and Hall, London, 1983.

Coward, D. A., G. S. Cripps, B. C. Maher, and B. H. Ridley, Zero magnetic field furnace controller, *1983 /1984 Biennial Rep.*, 26 pp., CSIRO Div. of Miner. Phys., North Ryde, New South Wales, Australia, 1985.

Cowie, J. W., and M. G. Bassett (Compilers), International Union of Geological Science 1989 Global Stratigraphic Chart, *Episodes, 12*(2), suppl., 1989.

Creer, K.M., A palaeomagnetic survey of South American rock formations, *Philos. Trans. R. Soc. London, Ser. A*, 267, 457-558, 1970.

Crowell, J. C., and L. A. Frakes, Late Palaeozoic glaciation of Australia, *J. Geol. Soc. Aust., 17*(2), 115-155, 1971.

Daly, L., and E. Irving, Paleomagnetisme des roches Carboniferes du Sahara central; analyse des aimantations juxtaposees; configuration de la Pangee, *Ann. Geophys., 1*(3), 207-216, 1983.

Fisher, R. A., Dispersion on a sphere, *Proc. R. Soc. London, Ser. A, 217*, 295-305, 1953.

Henry, B., N. Merabet, A. Yelles, M. S. Derder, and L. Daly, Geodynamical implications of a Moscovian palaeomagnetic pole from the stable Saharan craton (Illizi basin, Algeria), *Tectonophysics, 201*, 83-96, 1992.

Hurley, N. F., and R. Van der Voo, Paleomagnetism of Upper Devonian reefal limestones, Canning Basin, Western Australia, *Geol. Soc. Am. Bull., 98*, 138-146, 1987.

Idnurm, M., and C. A. Heinrich, A palaeomagnetic study of hydrothermal activity and uranium mineralization at Mt Painter, South Australia, *Aust. J. Earth Sci., 40*, 87-101, 1993.

Irving, E., Palaeomagnetism of some Carboniferous rocks from New South Wales and its relation to geological events, *J. Geophys. Res., 71*, 6025-6051, 1966.

Jackson, M., and R. Van der Voo, Drilling-induced remanence in carbonate rocks: occurrence, stability and grain-size dependence, *Geophys. J. R. Astron. Soc., 81*, 75-87, 1985.

Jesinkey, C., R. D. Forsythe, C. Mpodozis, and J. Davidson, Concordant Late Palaeozoic palaeomagnetizations for the Atacama Desert: Implications for tectonic models of the Chilean Andes, *Earth Planet. Sci. Lett., 85*, 461-472, 1987.

Kemp, E. M., B. E. Balme, R. J. Helby, R. A. Kyle, G. Playford, and P. L. Price, Carboniferous and Permian palynostratigraphy in Australia and Antarctica: A review, *BMR J. Aust. Geol. Geophys., 2*(3), 177-208, 1977.

Kent, D. V., O. Dia, and J. M. A. Sougy, Palaeomagnetism of Lower-Middle Devonian and Upper Proterozoic-Cambrian (?) rocks from Mejeria (Mauritania, West Africa), in *Plate Reconstruction From Palaeozoic Palaeomagnetism, Geodyn. Ser.*, vol. 12, edited by R. Van der Voo, C. R. Scotese, and N. Bonhommet, pp. 99-115, AGU, Washington, D.C., 1984.

Kent, D. V., and R. Van der Voo, Palaeozoic palaeogeography from palaeomagnetism of the Atlantic-bordering continents, in *Palaeozoic Palaeogeography and Biogeography, Mem. 12*, edited by W. S. McKerrow and C. R. Scotese, pp. 49-56, Geological Society, London, 1990.

Kirschvink, J. L., The least-squares line and plane and the analysis of palaeomagnetic data, *Geophys. J. R. Astron. Soc., 62*, 699-718, 1980.

Kligfield, R., W. Lowrie, A. Hirt, and A. W. B. Siddans, Effect of progressive deformation on remanent magnetization of Permian redbeds from the Alpes Maritimes (France), *Tectonophysics, 98*, 59-85, 1983.

Klootwijk, C., and J. Giddings, Palaeomagnetic results of Upper Palaeozoic volcanics, northeastern Queensland, and Australia's Late Palaeozoic APWP, in *NEO '93 Conference Proceedings*, pp. 617-627, University of New England, Armidale, New South Wales, Australia, 1993.

Kodama, K. P., Palaeomagnetism of granitic intrusives from the Precambrian basement under eastern Kansas: Orienting drill cores using secondary magnetization components, *Geophys. J. R. Astron. Soc., 76*, 272-287, 1984.

Li, Z. X., and C. M. Powell, Late Proterozoic to Early Paleozoic paleomagnetism and the formation of Gondwanaland, in *Gondwana 8; Assembly, Evolution and Dispersal*, edited by R. H. Findlay, M. R. Banks, J. J. Veevers and R. Unrug, pp.9-21, A. A. Balkema, Rotterdam, Netherlands, 1993.

Li, Z. X., P. W. Schmidt, and B. J. J. Embleton, Palaeomagnetism of the Hervey Group, central New South Wales and its tectonic implications, *Tectonics, 7*, 351-367, 1988.

Li, Z. X., C. M. Powell, and P. W. Schmidt, Syn-deformational remanent magnetization of the Mount Eclipse Sandstone, central Australia, *Geophys. J. Int., 99*, 205-222, 1989.

Li, Z. X., C. M. Powell, G. A. Thrupp, and P .W. Schmidt, Australian Palaeozoic palaeomagnetism and tectonics, II, A revised apparent polar wander path and palaeogeography, *J. Struct Geol., 12*(5/6), 567-575, 1990.

Li, Z. X., C. M. Powell, and D. G. Morris, Syn-deformational and drilling-induced remanent magnetizations from diamond drill cores of the Mt Eclipse Sandstone, central Australia, *Aust. J. Earth Sci., 38*, 473-484, 1991.

Li, Z. X., Z. Chen, and C. M. Powell, New Late Palaeozoic palaeomagnetic results from cratonic Australia, and revision of the Gondwanan apparent polar wander path, *Explor. Geophys.*, in press, 1993.

Martin, D. L., A. E. M. Naim, H. C. Noltimier, M. H. Petty, and T. J. Schmitt, Paleozoic and Mesozoic paleomagnetic results from Morocco, *Tectonophysics, 44*, 91-114, 1978.

McElhinny, M. W., *Palaeomagnetism and Plate Tectonics*, 358 pp., Academic, San Diego, Calif., 1973.

McElhinny, M. W., and B. J. J. Embleton, The palaeoposition of Madagascar: Remanence and magnetic properties of Late Palaeozoic sediments, *Earth Planet. Sci. Lett., 31*, 101-112, 1976.

McElhinny, M. W., and N. D. Opdyke, Paleomagnetism of some Carboniferous glacial varves from Central Africa, *J. Geophys. Res., 73*, 689-696, 1968.

McFadden, P. L., A new fold test for palaeomagnetic studies, *Geophys. J. Int., 103*, 163-169, 1990.

McFadden, P. L., and M. W. McElhinny, Classification of the reversal test in palaeomagnetism, *Geophys. J. Int., 103*, 725-729, 1990.

McWilliams, M., and M. J. Pinto, Paleomagnetic results from granitic basement rocks in the Cajon Pass scientific drillhole, *Geophys. Res. Lett., 15*(9), 1069-1072, 1988.

Meert, J. G., R. Van der Voo, C. M. Powell, Z. X. Li, M. W. McElhinny, and D. T. A. Symons, A plate-tectonic speed limit? *Nature, 363*, 216-217, 1993.

Mortimer, G. E., J. A. Cooper, and P. R. James, U-Pb and Rb-Sr geochronology and geological evolution of the Harts Range ruby mine area of the Arunta Inlier, central Australia., *Lithos, 20*, 445-467, 1987.

Opdyke, N. D., The paleomagnetism of the Permian redbeds of southwest Tanganyika, *J. Geophys. Res , 69*, 2477-2487, 1964.

Playford, G., Lower Carboniferous spores from the Ducabrook Formation, Drummond Basin, Queensland, *Palaeontograph. Abt., Ser. B, 167*, 105-160, 1978.

Playford, G., Palynology of the Australian Lower Carboniferous: A review, in *Compte Rendu, 10e Congrés International de Stratigraphie et de Géologie du Carbonifére, Màdrid, Spain 1983*, vol 4, pp. 247-265, Graficas San Martin, Màdrid, 1985.

Powell, C. M, and J. J. Veevers, Namurian uplift in Australia and South America triggered the main Gondwanan glaciation, *Nature, 326*, 177-179, 1987.

Powell, C. M, S. R. Roots, and J. J. Veevers, Pre-breakup continental extension in East Gondwanaland and the early opening of the eastern Indian Ocean, *Tectonophysics, 155*, 261-283, 1988.

Rapalini, A. E., and J. F. Vilas, Tectonic rotations in the Late Palaeozoic continental margins of southern South America determined and dated by palaeomagnetism, *Geophys. J. Int., 107*, 333-351, 1991.

Runnegar, B., and K. S. W. Campbell, Late Palaeozoic Faunas of Australia, *Earth Sci. Rev., 12*, 235-257, 1976.

Saucier, A. E., Uranium in the Ngalia Basin, Northern Territory, Australia, report , AGIP Australia Pty Ltd., Sydney, Australia, 1981.

Schmidt, P. W., Linearity spectrum analysis of multi-component magnetizations and its application to some igneous rocks from south-eastern Australia, *Geophys. J. R. Astron. Soc., 70*, 647-665, 1982.

Schmidt, P. W., B. J. J. Embleton, and H. C. Palmer, Pre- and post-folding magnetization from the Devonian Snowy River Volcanics and Buchan Caves Limestone, Victoria, *Geophys. J. R. Astron. Soc., 91*, 155-170, 1987.

Schmidt, P. W., C. M. Powell, Z. X. Li, and G. A. Thrupp, Reliability of Palaeozoic palaeomagnetic poles and APWP of Gondwanaland, *Tectonophysics, 184*, 87-100, 1990.

Sclater, J. G., and R. L. Fischer, Evolution of the east central Indian Ocean with emphasis on the tectonic setting of the Ninetyeast Ridge, *Geol. Soc. Am. Bull., 85*, 683-702, 1974.

Scotese, C. R., and S. F. Barrett, Gondwana's movement over the South Pole during the Paleozoic: Evidence from lithologic indicators of climate, in *Palaeozoic Palaeogeography and Biogeography, Mem. 12*, edited by W. S. McKerrow and C. R. Scotese, pp.75-85, Geological Society, London, 1990.

Scotese, C. R., and W. S. McKerrow, Revised world maps and introduction, in *Palaeozoic Palaeogeography and Biogeography, Mem. 12*, edited by W. S. McKerrow and C. R. Scotese, pp. 1-21, Geological Society, London, 1990.

Shaw, R. D., and L. P. Black, The history of the Redbank Thrust Zone, central Australia, based on structural, metamorphic and Rb-Sr isotopic evidence: Tectonic implications, *Aust. J. Earth Sci , 38*, 307-332, 1991.

Sinito, A. M., D. A. Valencio, and J. F. Vilas, Palaeomagnetism of a sequence of Upper Palaeozoic–Lower Mesozoic red beds from Argentina, *Geophys. J. R. Astron. Soc., 58*, 237-247, 1979.

Soffel, H. C., S. Saradeth, J. C. Briden, V. Bachtadse, and C. Rolf, The Sabaloka ring complex revisited: Palaeomagnetism and rockmagnetism, *Geophys. J. Int , 102*, 411-420, 1990.

Thrupp, G. A., D. V. Kent, P. W. Schmidt, and C. M. Powell, Palaeomagnetism of red beds of the Late Devonian Worange Point Formation, SE Australia, *Geophys. J Int., 104*, 179-201, 1991.

Van der Voo, R., Phanerozoic paleomagnetic poles from Europe and North America and comparisons with continental reconstructions, *Rev. Geophys., 28*, 167-206, 1990.

Veevers, J. J., and C. M. Powell, Late Palaeozoic glacial episodes in Gondwanaland reflected in transgressive-regressive depositional sequences in Euramerica, *Geol. Soc. Am. Bull., 98*, 475-487, 1987.

Wells, A. T., and F. J. Moss, The Ngalia Basin, Northern Territory: Stratigraphy and structure, *BMR. Bull., 212*, 88 pp., 1983.

Wensink, H., and C. Klootwijk, The paleomagnetism of the Talchir Series of the Lower Gondwana System, Central India, *Earth Planet Sci. Lett., 4*, 191-196, 1968.

Young, G.C., Devonian vertebrate distribution patterns and cladistic analysis of palaeogeographic hypotheses, in *Palaeozoic Palaeogeography and Biogeography, Mem 12*, edited by W. S. McKerrow and C. R. Scotese, pp. 243-255, Geological Society, London, 1990.

Zijderveld, J. D. A., A. C. demagnetization of rocks: analysis of results, in *Methods in Palaeomagnetism*, pp. 254-286, edited by D. W. Collinson, K. M. Creer and S. K. Runcorn, Elsevier, New York, 1967.

B. E. Balme, Z. Chen, Z. X. Li, C. M. Powell, Department of Geology, University of Western Australia, Nedlands, WA 6009, Australia. (e-mail: z_li@fennel.cc.uwa.oz.au)

(Received November 2, 1992; revised June 29, 1993; accepted August 6, 1993.)